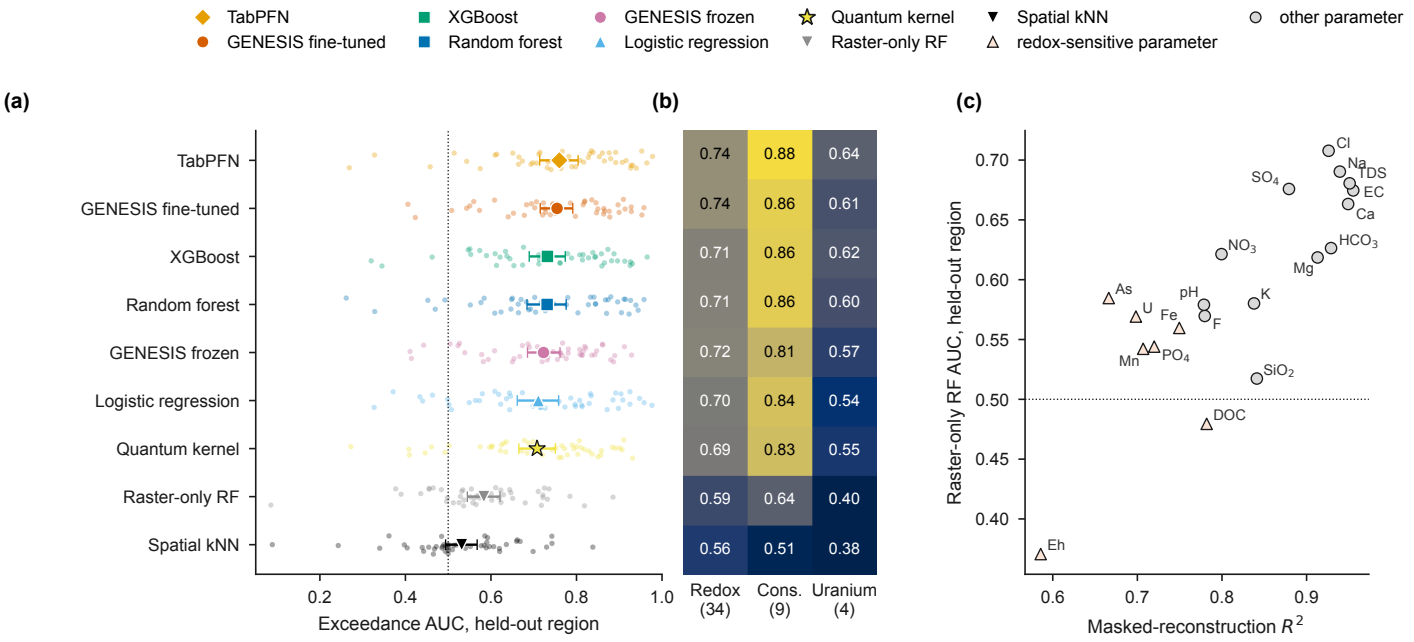

TabPFN
GENESIS fine-tuned
XGBoost
Random forest
GENESIS frozen
Logistic regression
Quantum kernel
Raster-only RF
Spatial kNN
redox-sensitive parameter
other parameter
(a)
(b)
(c)
TabPFN
GENESIS fine-tuned
XGBoost
Random forest
GENESIS frozen
Logistic regression
Quantum kernel
Raster-only RF
Spatial kNN
0.2
0.4
0.6
0.8
1.0
Exceedance AUC, held-out region
0.74 0.88 0.64
0.74 0.86 0.61
0.71 0.86 0.62
0.71 0.86 0.60
0.72 0.81 0.57
0.70 0.84 0.54
0.69 0.83 0.55
0.59 0.64 0.40
0.56 0.51 0.38
Redox (34)
Cons. (9)
Uranium (4)
Raster-only RF AUC, held-out region
0.70
0.65
0.60
0.55
0.50
0.45
0.40
Cl
Na
TDS
EC
Ca
SO4
NO3
HCO3
Mg
K
As
pH
U
Fe
F
Mn
PO4
SiO2
DOC
Eh
0.6
0.7
0.8
0.9
Masked-reconstruction R^2